Sentinel-2 Detection of Urban Expansion Hotspots in the Petah Tikva Urban Fringe (2018–2025)

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1. Introduction

Urban expansion has become an increasingly critical topic worldwide over the last few decades. It touches on key issues such as housing, transportation, water provision, and environmental compliance. The Gush Dan metropolitan region in central Israel is a stark showcase of these pressures, being economically dynamic while simultaneously experiencing rapid population growth and urban land-use conversion.

A number of previous scientific investigations were consulted for the conduction of this research.

- Arif and Toersilowati (2024) monitored and predicted the growth of built-up areas in suburban Yogyakarta, Indonesia using spectral metrics.
- Nandi, Sahu and Beuria (2026) analysed land cover transformation in Bhubaneswar, India using Landsat satellite imagery and supervised machine learning classification.
- Chai et al. (2025) performed time-series analysis on urban green space in Shanghai, China based on Sentinel-2 data.
- Close et al. (2021) investigated the feasibility of using Sentinel-2 for land cover change analysis in Wallonia, Belgium.
- Haxhiu and Aliaj (2025) researched the use of remote sensing and open geospatial data for detecting urban change trends in Tirana, Albania.
- Herbei et al. (2026) performed AI-supported detection of vegetation degradation and urban expansion in Timișoara, Romania using Sentinel-2 data and Random Forest (RF).
- Jeong (2024) monitored land cover changes using time-series Sentinel-2 imagery analysed with Support Vector Machine (SVM).
- Karlsen et al. (2021) used time-series cloud-free NDVI data to map the onset of vegetation growth in Svalbard, Norway.
- Kopec, Wijata and Nalepa (2025) performed change detection on Sentinel-2 images using deep learning ensembles.
- Lefebvre, Sannier and Corpetti (2016) proposed exploiting Sentinel-2 imagery to update the Copernicus High Resolution Layer imperviousness dataset.
- Tempa et al. (2024) utilized Sentinel-2 imagery and Random Forest (RF) for analysing land cover and NDVI change dynamics in Gelephu, Bhutan.

Taking insights from previous research, this investigation aims to analyse land-use change trends between 2018 and 2025 along the southeastern edge of the city of Petah Tikva, focusing on a sharp urban–rural transition zone. The study uses temporal satellite data, including spectral bands and derived indices, to identify urban conversion sites over the eight-year period.

Google Earth Engine and Python are used to process and analyse the data, enabling the identification of major urban change hotspots and the quantification of changes in key spectral indices: NDVI, NDBI, and BSI, representing vegetation cover, built-up surfaces, and bare ground respectively over the study period.

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2. Method

The goal of this investigation was to determine land cover change trends within the Petah Tikva area using satellite data, specifically the NDVI (Normalized Difference Vegetation Index), NDBI (Normalized Difference Built-up Index), and BSI (Bare Soil Index), combined with hotspot analysis.

NDVI describes the vegetation density of an area and is commonly used to assess changes in plant health (GISRSStudy, 2026). It is calculated as a ratio between the red and near-infrared (NIR) spectral bands. NDBI describes the built-up density of an area and is used to assess trends in urban development, calculated as a ratio between shortwave infrared (SWIR) and NIR bands. BSI describes the occurrence of bare soil and is used to distinguish exposed ground from other land cover types, calculated using SWIR, red, and blue spectral bands (ClimateEngine.org, 2025).

Hotspot analysis is a spatial analysis and mapping technique used to identify clustering of spatial phenomena (Columbia University Mailman School of Public Health, 2026). In this investigation, hotspots represent spatial areas—primarily along the southeastern edge of Petah Tikva—where the largest extent of urban conversion was observed between 2018 and 2025.

Two main tools were used. Google Earth Engine (GEE) was used to define the area of interest (AOI), select Sentinel-2 imagery, calculate NDVI, NDBI, and BSI, and export both spectral bands and derived indices as raster datasets. Python was then used to compute index changes between 2018 and 2025, define urban conversion and construction masks, remove noise, extract hotspot polygons, and visualize the results.

The analysis followed these steps:

- A temporal range from 2018 to 2025 was selected
- An AOI covering the southeastern Petah Tikva fringe was defined
- The Copernicus Harmonized Sentinel-2 dataset was selected for its high-resolution and consistent coverage

- A dry-season window (May–July) was used to improve visibility of built surfaces
- Annual composites (2018, 2021, 2023, 2025) were generated and visualized
- Spectral bands (B2, B3, B4, B8, B11, B12) and indices (NDVI, NDBI, BSI) were exported

In Python:

- Raster datasets were imported and bands assigned
- Change layers (ΔNDVI, ΔNDBI, ΔBSI) between 2018 and 2025 were calculated
- A rule-based threshold approach was used to define:
    - urban conversion mask
    - construction-stage mask
- Small noisy pixel clusters were removed using Rasterio-based filtering
- Cleaned conversion masks were saved as GeoTIFFs
- Total changed area was calculated from pixel counts and converted to hectares
- Raster masks were converted to polygons using GeoPandas and Shapely
- The largest hotspot polygons were identified and ranked by area
- Visual outputs were generated, including:
    - cleaned conversion masks
    - hotspot overlays on NDVI and BSI
    - histogram distributions of index changes for threshold evaluation

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3. Results

The analysis identified significant land cover change between 2018 and 2025.

- Likely vegetation/open-to-built conversion: **379.88 ha**
- Likely construction-stage land: **551.21 ha**
- Candidate conversion pixels: **46,063**
- Cleaned conversion pixels: **37,988**

The largest hotspot polygons ranged between approximately **7.96 ha and 25.69 ha**, indicating substantial localized land transformation:

Rank	Area (ha)
1	25.69
2	24.81
3	20.60
4	18.11
5	17.44
6	16.60
7	9.86

Rank	Area (ha)
8	9.57
9	8.77
10	7.96

The spectral change metrics within detected hotspots showed strong and consistent patterns:

- Mean ΔNDVI: **-0.4647**
- Mean ΔNDBI: **+0.2939**
- Mean ΔBSI (construction areas): **+0.2692**

These results indicate:

- a large decrease in vegetation
- a significant increase in built-up surfaces
- a notable increase in bare soil, consistent with active construction

Spatial outputs showed:

- clear clustering of transformation zones
- strong alignment between detected hotspots and urban fringe expansion
- consistent index-based signals across multiple visualization layers

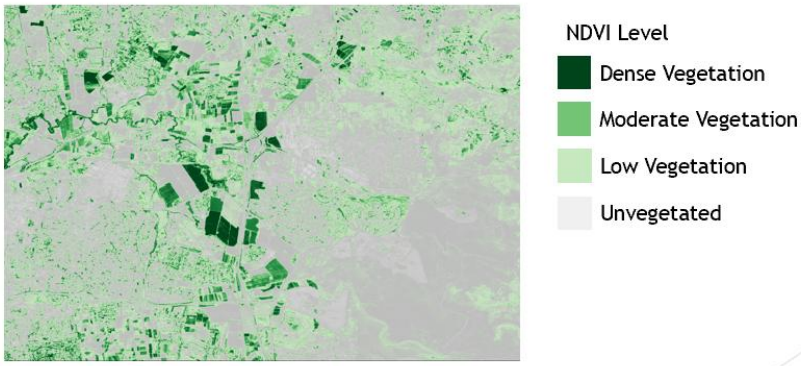

Figure 1: Petah Tikva NDVI level 2018

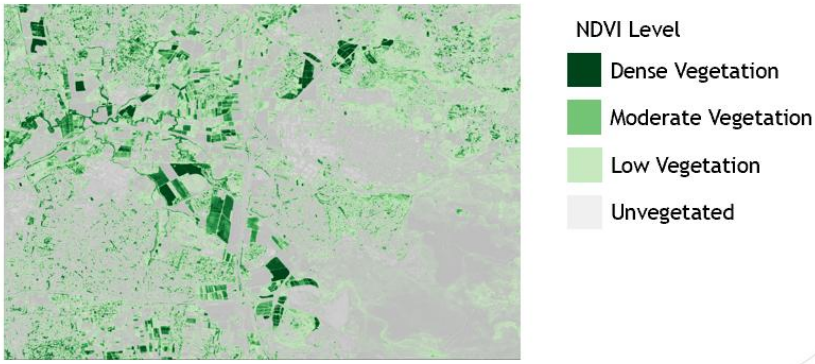

Figure 2: Petah Tikva NDVI cover level 2025

Figure 3: Cleaned conversion mask between 2018 and 2025

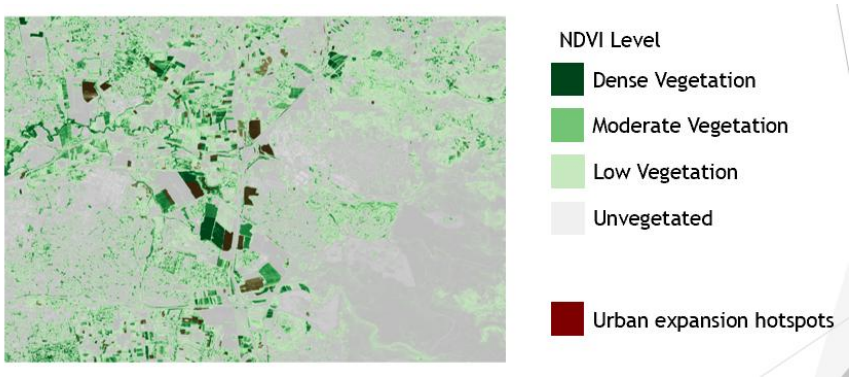

Figure 4: Likely urban conversion hotspots over 2018 NDVI

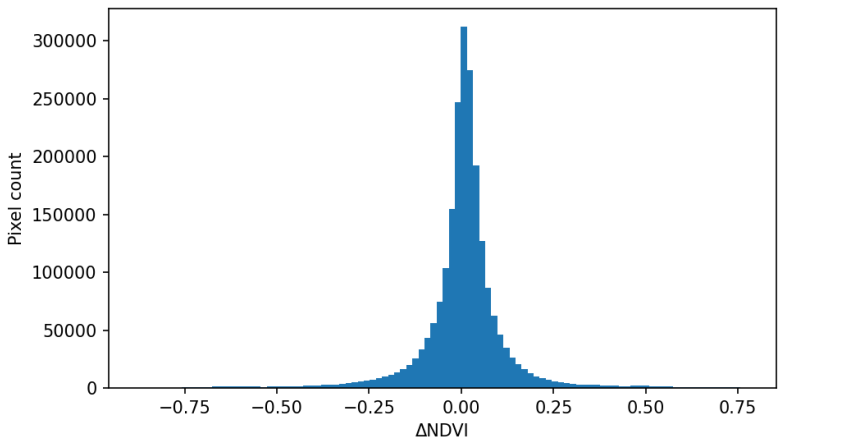

Figure 5: Distribution of NDVI between 2018 and 2025

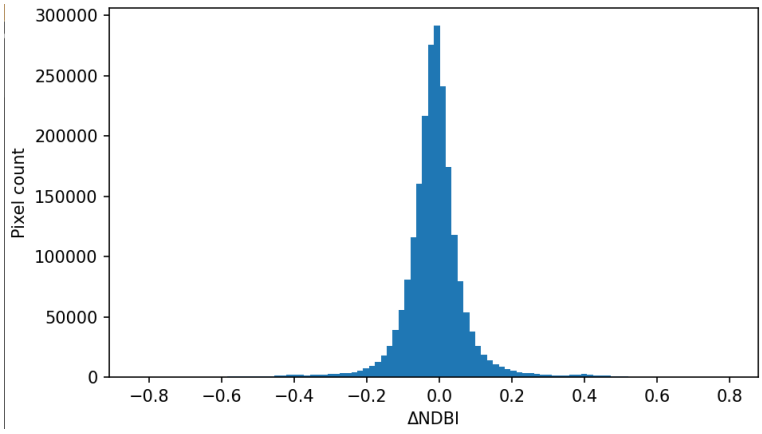

Figure 6: Distribution of NDBI between 2018 and 2025

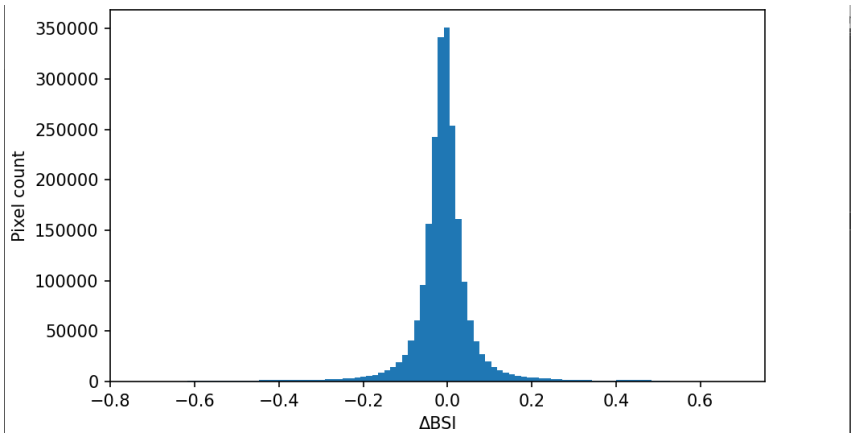

Figure 7: Distribution of BSI between 2018 and 2025

4. Discussion

Most urban conversion hotspots are clustered in a south-easterly direction, reflecting recent urban expansion trends in the Petah Tikva fringe. The results show a clear and consistent pattern: NDVI decreases substantially within transformation hotspots, while NDBI and BSI increase. This indicates a marked reduction in vegetation over the seven-year period, accompanied by the progressive transition to bare soil and built surfaces.

The magnitude of the observed index changes suggests that the detected transformations are not driven by seasonal variability or noise, but rather represent genuine structural land cover change. The combined behaviour of ΔNDVI, ΔNDBI, and ΔBSI reflects the expected sequence of urban development: vegetation clearance, exposure of soil during construction, and eventual built-up expansion.

A technical limitation encountered during the investigation was the incompatibility of the skimage library with Python 3.14, which required replacing the small-object removal step with a Rasterio-based approach. Despite this adjustment, the cleaning process remained effective and produced coherent spatial outputs.

A refinement was applied to the visualization of the Bare Soil Index (BSI) to improve interpretability and avoid potential misrepresentation of surface conditions. The default terrain colour scheme was found to introduce blue tones at low BSI values, which may be visually associated with water bodies or unrelated spectral responses and could therefore lead to confusion in the context of urban land transformation analysis.

To address this, a custom colour gradient ranging from green (low BSI) through yellow to brown (high BSI) was implemented, aligning more closely with the physical meaning of the index, where increasing values correspond to greater exposure of bare soil and disturbed surfaces.

This adjustment enhances the clarity of the results by providing a more intuitive visual distinction between vegetated, transitional and construction-stage areas, and ensures consistency with the interpretation of NDVI and NDBI changes across the study area.

This investigation demonstrates strong applicability for identifying hotspots of recent urban expansion in transitional peri-urban zones. The methodology is transferable to other regions with reliable Sentinel-2 coverage and can be extended further. Future work could incorporate:

- temporal trend analysis across more years
- integration with ancillary datasets (e.g. roads, zoning)
- machine learning approaches to predict future expansion patterns (Gimmler and Lange, 2021)

Overall, the approach provides a robust, scalable framework for monitoring land cover change and identifying emerging urban growth zones.

5. Conclusion

Like many other major urban areas around the world, Gush Dan is growing rapidly in both extent and population, placing increasing strain on local natural resources and infrastructure and highlighting the need for informed urban planning and monitoring.

This investigation analysed urban land conversion along the southeastern edge of Petah Tikva at the urban–rural interface using Sentinel-2 satellite imagery, spectral bands, and derived indices. The results indicate a notable increase in the transformation of land to urban use between 2018 and 2025, with a clear spatial trend in a southeasterly direction.

A limitation encountered during the investigation was the incompatibility of the skimage library with the development environment, which required the use of an alternative method for removing small noisy pixel groups. Despite this, the overall workflow remained effective and produced coherent results.

The methodology applied in this study can be further expanded by incorporating a longer temporal analysis, integrating satellite imagery with ancillary datasets such as road networks and land-use zoning, and applying machine learning approaches to predict future urban expansion patterns.

Overall, this investigation presents a clear and practical method for identifying hotspots of recent urban land conversion along urban–rural interfaces.

6. References

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7. Code Implementation

7.1 Google Earth Engine (GEE): Satellite Imagery, Spectral Indices and Export

```
import rasterio

import numpy as np

from pathlib import Path

from rasterio.features import sieve, shapes

from shapely.geometry import shape

import geopandas as gpd

import matplotlib.pyplot as plt

from matplotlib.colors import LinearSegmentedColormap, ListedColormap


# --------------------------------------------------

# Add the .tif file paths

# --------------------------------------------------

data_dir = Path("data")


files = {

    2018: data_dir / "PetahTikva_S2_2018.tif",

    2021: data_dir / "PetahTikva_S2_2021.tif",

    2023: data_dir / "PetahTikva_S2_2023.tif",

    2025: data_dir / "PetahTikva_S2_2025.tif",

}


def read_stack(path):

    with rasterio.open(path) as src:

        arr = src.read()  # shape: (bands, rows, cols)

        profile = src.profile

    return arr, profile
```

```
stacks = {}
profiles = {}

for year, path in files.items():
    arr, profile = read_stack(path)
    stacks[year] = arr
    profiles[year] = profile

# ---------------------------------------------------
# Create a band index
# ---------------------------------------------------
BAND_INDEX = {
    "B2": 0,
    "B3": 1,
    "B4": 2,
    "B8": 3,
    "B11": 4,
    "B12": 5,
    "NDVI": 6,
    "NDBI": 7,
    "BSI": 8,
}

# ---------------------------------------------------
# Compute the change layers (layer differences)
# ---------------------------------------------------
def diff_band(year1, year2, band_name):
    i = BAND_INDEX[band_name]
    return stacks[year2][i] - stacks[year1][i]

d_ndvi_18_25 = diff_band(2018, 2025, "NDVI")
d_ndbi_18_25 = diff_band(2018, 2025, "NDBI")
```

```
d_bsi_18_25  = diff_band(2018, 2025, "BSI")

# ---------------------------------------------------
# Create a rule-based urban conversion mask
# ---------------------------------------------------
ndvi_2018 = stacks[2018][BAND_INDEX["NDVI"]]
ndvi_2025 = stacks[2025][BAND_INDEX["NDVI"]]
ndbi_2018 = stacks[2018][BAND_INDEX["NDBI"]]
ndbi_2025 = stacks[2025][BAND_INDEX["NDBI"]]
bsi_2025  = stacks[2025][BAND_INDEX["BSI"]]

candidate_conversion = (
    (ndvi_2018 > 0.35) &               # was vegetated/open
    (ndvi_2025 < 0.30) &               # vegetation declined
    (d_ndvi_18_25 < -0.15) &           # clear NDVI drop
    (d_ndbi_18_25 > 0.10) &            # built-up signal rose
    (ndbi_2025 > 0.00)                 # now more built-like
)

# Detect likely construction zones
candidate_construction = (
    (d_ndvi_18_25 < -0.15) &
    (d_bsi_18_25 > 0.10) &
    (bsi_2025 > 0.05)
)

# ---------------------------------------------------
# Remove very small isolated patches
# size=20 means remove connected patches smaller than 20 pixels
# ---------------------------------------------------
conversion_clean = sieve(candidate_conversion.astype("uint8"),
size=20).astype(bool)
```

```
construction_clean = sieve(candidate_construction.astype("uint8"),
size=20).astype(bool)


print("Conversion pixels after cleaning:", conversion_clean.sum())

print("Construction pixels after cleaning:", construction_clean.sum())


# --------------------------------------------------

# Save change masks as GeoTIFF

# --------------------------------------------------

BASE_DIR = Path(__file__).resolve().parent

outputs_dir = BASE_DIR / "outputs"

outputs_dir.mkdir(exist_ok=True)


def write_single_band_geotiff(output_path, array, reference_profile,
dtype="uint8"):

    profile = reference_profile.copy()

    profile.update(

        count=1,

        dtype=dtype,

        compress="lzw"

    )


    with rasterio.open(output_path, "w", **profile) as dst:

        dst.write(array.astype(dtype), 1)


write_single_band_geotiff(

    outputs_dir / "conversion_2018_2025.tif",

    conversion_clean,

    profiles[2018]

)


write_single_band_geotiff(

    outputs_dir / "construction_2018_2025.tif",
```

```
    construction_clean,
    profiles[2018]
)

print("Change masks saved successfully.")

# ---------------------------------------------------
# Save cleaned conversion mask as PNG (black and white)
# ---------------------------------------------------
plt.figure(figsize=(8, 8))
plt.imshow(conversion_clean, cmap="gray", vmin=0, vmax=1)
plt.title("Cleaned conversion mask (2018–2025)")
plt.axis("off")
plt.savefig(
    outputs_dir / "conversion_2018_2025_mask.png",
    bbox_inches="tight",
    pad_inches=0,
    dpi=300
)
plt.show()

print("Black-and-white PNG mask saved successfully.")

# ---------------------------------------------------
# Calculate changed areas and convert to hectares
# ---------------------------------------------------
pixel_area_m2 = 10 * 10  # Sentinel-2 10 m pixel

conversion_area_m2 = conversion_clean.sum() * pixel_area_m2
construction_area_m2 = construction_clean.sum() * pixel_area_m2

conversion_area_ha = conversion_area_m2 / 10000
```

```
construction_area_ha = construction_area_m2 / 10000

print(f"Likely vegetation/open-to-built conversion: {conversion_area_ha:.2f} ha")
print(f"Likely construction-stage land: {construction_area_ha:.2f} ha")

# --------------------------------------------------
# Load the same raster and convert it to polygons
# --------------------------------------------------
def raster_mask_to_polygons(mask, transform, crs):
    geoms = []

    for geom, value in shapes(mask.astype("uint8"), mask=mask,
transform=transform):
        if value == 1:
            geoms.append(shape(geom))

    gdf = gpd.GeoDataFrame(geometry=geoms, crs=crs)

    if not gdf.empty:
        gdf = gdf.to_crs(epsg=2039)
        gdf["area_m2"] = gdf.geometry.area
        gdf["area_ha"] = gdf["area_m2"] / 10000
        gdf = gdf.sort_values("area_m2", ascending=False).reset_index(drop=True)

    return gdf

print("Candidate conversion pixels:", candidate_conversion.sum())
print("Cleaned conversion pixels:", conversion_clean.sum())

# Create hotspots variable with geospatial table
with rasterio.open(outputs_dir / "conversion_2018_2025.tif") as src:
    conv_mask = src.read(1).astype(bool)
    transform = src.transform
```

```
    crs = src.crs

hotspots = raster_mask_to_polygons(conv_mask, transform, crs)

print(hotspots.head(10))
print("Number of hotspot polygons:", len(hotspots))

# Save hotspot polygons
if not hotspots.empty:
    hotspots.to_file(outputs_dir / "conversion_hotspots.geojson",
driver="GeoJSON")
    print("Hotspots saved as GeoJSON.")
else:
    print("No hotspot polygons found.")

# Show top hotspot patches
if not hotspots.empty:
    print("\nTop 10 largest conversion hotspots:")
    print(hotspots[["area_ha"]].head(10))
else:
    print("No hotspots to display.")

# ---------------------------------------------------
# Colormaps
# ---------------------------------------------------

# NDVI colormap matching the GEE NDVI palette
ndvi_colors = [
    "#f0f0f0",  # light gray
    "#d9d9d9",
    "#bdbdbd",
    "#c7e9c0",  # pale green
    "#74c476",  # medium green
```

```
    "#238b45",
    "#00441b"   # dark green
]
ndvi_cmap = LinearSegmentedColormap.from_list("gee_ndvi", ndvi_colors, N=256)

# Harmonized hotspot overlay color
hotspot_cmap = ListedColormap(["#7f0000"])

# Fixed BSI colormap: green -> yellow -> brown
bsi_colors = [
    "#1a9850",  # vegetation / low BSI
    "#91cf60",
    "#d9ef8b",
    "#fee08b",
    "#fdae61",
    "#a6611a",
    "#6b3d0c"   # strong bare soil / high BSI
]
bsi_cmap = LinearSegmentedColormap.from_list("bsi_clean", bsi_colors, N=256)

# ---------------------------------------------------
# Plot the cleanest conversion mask
# ---------------------------------------------------
plt.figure(figsize=(8, 8))
plt.imshow(conversion_clean, cmap="gray", vmin=0, vmax=1)
plt.title("Cleaned conversion mask (2018–2025)")
plt.axis("off")
plt.show()

# ---------------------------------------------------
# Plot the conversion mask over the NDVI background
# ---------------------------------------------------
```

```
plt.figure(figsize=(8, 8))
plt.imshow(ndvi_2018, cmap=ndvi_cmap, vmin=-0.2, vmax=0.8)
plt.imshow(
    np.ma.masked_where(~conversion_clean, conversion_clean),
    cmap=hotspot_cmap,
    alpha=0.45,
    vmin=0,
    vmax=1
)
plt.title("Likely urban conversion hotspots over 2018 NDVI")
plt.axis("off")
plt.savefig(
    outputs_dir / "ndvi2018_hotspots_overlay.png",
    bbox_inches="tight",
    pad_inches=0,
    dpi=300
)
plt.show()

print("NDVI 2018 hotspot overlay PNG saved successfully.")

# ---------------------------------------------------
# Plot the construction mask over the BSI background
# ---------------------------------------------------
plt.figure(figsize=(8, 8))
plt.imshow(bsi_2025, cmap=bsi_cmap, vmin=-0.3, vmax=0.4)
plt.imshow(
    np.ma.masked_where(~construction_clean, construction_clean),
    cmap=hotspot_cmap,
    alpha=0.55,
    vmin=0,
    vmax=1
```

```
)
plt.title("Likely construction-stage areas over 2025 BSI")
plt.axis("off")
plt.savefig(
    outputs_dir / "bsi2025_construction_overlay.png",
    bbox_inches="tight",
    pad_inches=0,
    dpi=300
)
plt.show()

print("BSI 2025 construction overlay PNG saved successfully.")

# ---------------------------------------------------
# Plot index-change histograms for threshold tuning
# ---------------------------------------------------
plt.figure(figsize=(7, 4))
plt.hist(d_ndvi_18_25.ravel(), bins=100)
plt.title("Distribution of ΔNDVI (2018–2025)")
plt.xlabel("ΔNDVI")
plt.ylabel("Pixel count")
plt.show()

plt.figure(figsize=(7, 4))
plt.hist(d_ndbi_18_25.ravel(), bins=100)
plt.title("Distribution of ΔNDBI (2018–2025)")
plt.xlabel("ΔNDBI")
plt.ylabel("Pixel count")
plt.show()

plt.figure(figsize=(7, 4))
plt.hist(d_bsi_18_25.ravel(), bins=100)
```

```
plt.title("Distribution of ΔBSI (2018–2025)")

plt.xlabel("ΔBSI")

plt.ylabel("Pixel count")

plt.show()


# ---------------------------------------------------

# Determine NDVI, NDBI and BSI trends

# ---------------------------------------------------

print("Mean ΔNDVI in hotspots:", d_ndvi_18_25[conversion_clean].mean())

print("Mean ΔNDBI in hotspots:", d_ndbi_18_25[conversion_clean].mean())

print("Mean ΔBSI in construction hotspots:",
d_bsi_18_25[construction_clean].mean())
```